

Relieve hotspots

When overflow appears, global routers rip up contributing nets and reroute around congested edges

The idea

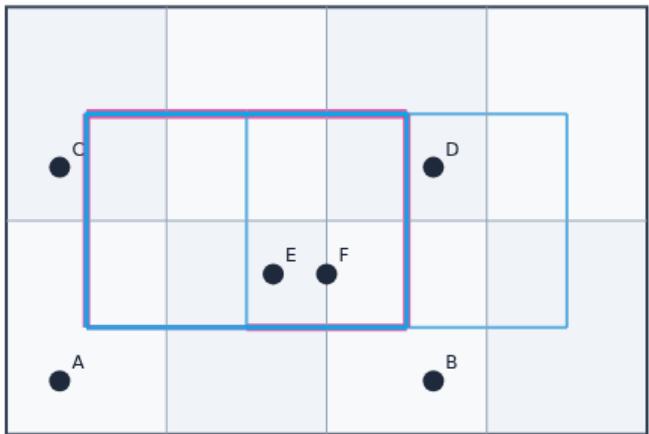
- Score each net by overflow on its edges
- Subtract its route from usage
- Run `maze_route` between its pins with the remaining usage map
- Add the new edges back
- Total overflow should not rise; ideally it drops versus the pre-rip state

Sequential L first

RIP-UP REROUTE

Sequential L first

Step 1 / 5 · seq · module03-03-ripup-reroute



Explanation

Route L-HV; some edges exceed Cap.

- Overflow appears
- Before rip

total≈2

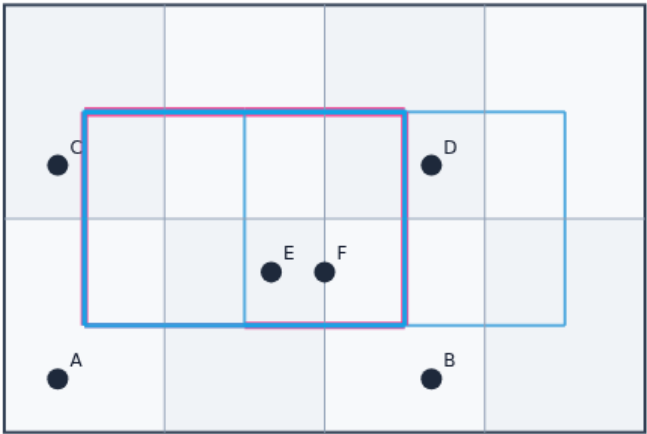
Sequential L first

Pick hot edge

RIP-UP REROUTE

Pick hot edge

Step 2 / 5 · pick · module03-03-ripup-reroute



Explanation

Find worst overflowing edge; mark nets using it.

- Shared corridor
- Toy rip order

Worst edge first

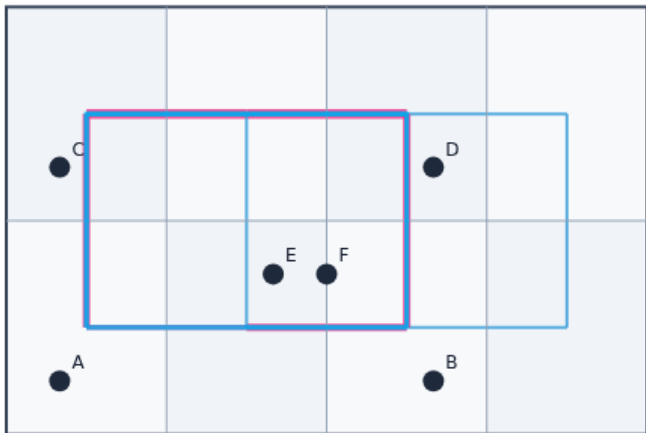
Pick hot edge

Rip nets

RIP-UP REROUTE

Rip nets

Step 3 / 5 · rip · module03-03-ripup-reroute



Explanation

Remove those nets from usage.

- Subtract paths
- Ready to reroute

Usage drops

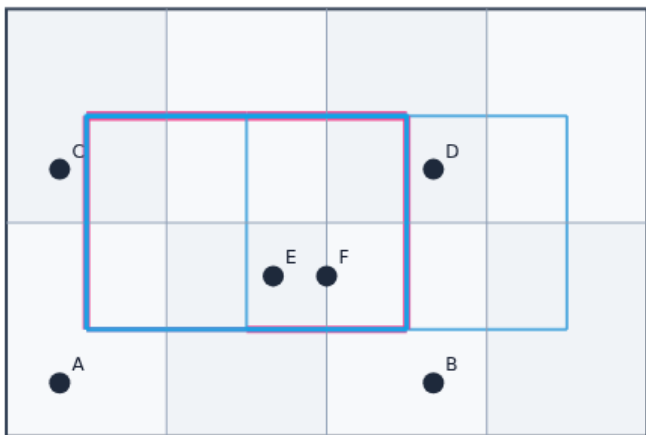
Rip nets

Maze reroute

RIP-UP REROUTE

Maze reroute

Step 4 / 5 · maze-r · module03-03-ripup-reroute



Explanation

Ripped nets maze-route with updated usage.

- May improve
- GOLDENS.ripupImprove

After ripup

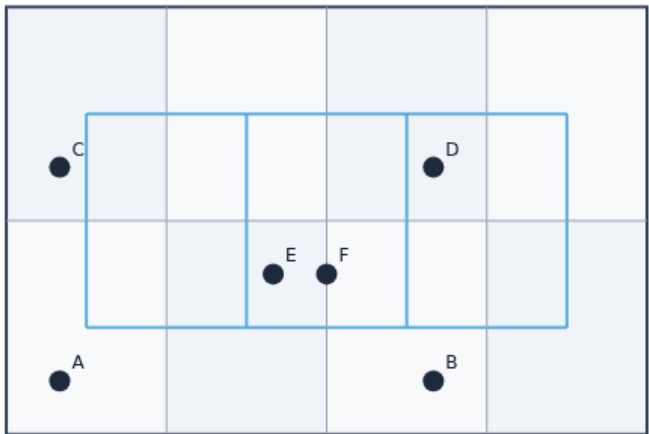
Maze reroute

Iterate

RIP-UP REROUTE

Iterate

Step 5 / 5 · loop · module03-03-ripup-reroute



Explanation

Real routers loop estimate → route → rip until clean or budget.

- Detailed route next
- learn_routing

Course arc

Iterate

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / Rip-up and reroute

INTERACTIVE TOOL

Rip-up and reroute

After sequential L routing overflows, rip-up and maze-reroute hot nets.

Your job: Route L-HV, note overflow, click Rip-up reroute. Golden: rip-up can reduce total on hot corridors.

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] Route L-HV first

Pick a challenge and click **Start**. Move cells, route, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A

B

C

D

E

F

←

→

↑

↓

Reset to starter

Reveal golden (study)

Route L-HV

Route L-VH

Route maze

Rip-up reroute

Graph

Reference / metrics

selected: A @ GCell (0,0)
routes: (none — click Route)
overflow total: 0.00 · max: 0.00 · count: 0
edge capacity: 2
HPWL: 34.0
view: your placement

Browser lab starter

learn_global_routing — 03 ripup reroute

Implement track

- Implement ``ripup_reroute(routes, usage, capacity, nets, term, nx, ny)``
- Assert total overflow after is less than or equal to before on `tiny_gr` sequential L seed

Pitfalls

- Ripping a net but leaving ghost usage on its old edges
- Rerouting with pattern L through the same hot edge
- Picking the wrong net to rip, use overflow contribution not HPWL

Your turn

- Clear rip-up challenges
- Next: tie it together with full sequential global routing